

Ring Proof Specification

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14-03-2026 Draft-8

Abstract

This document specifies the SNARK component originally designed as part of the Ring VRF construction [1]. The scheme enables a prover to demonstrate knowledge of a secret scalar t and a secret index k within a group of public keys, where each public key is a point on an elliptic curve. The scheme ensures that, when combined with a public elliptic curve point H , the relation $R = PK_k + t \cdot H$ is satisfied. It leverages elliptic curve operations, a polynomial commitment scheme, and the Fiat-Shamir heuristic to achieve non-interactivity and zero-knowledge properties.

1. Notation

1.1. Basics

Basic Sets

- $\mathbb{N}_k = \{0, \dots, k-1\}$
- $\mathbb{B} = \mathbb{N}_2$

Vectors Operations

- $\bar{x} = (x_0, \dots, x_{n-1})$, $\bar{x}_i = x_i$, $0 \leq i < n$
- $\bar{a} \parallel \bar{b} = (a_0, \dots, a_{n-1}, b_0, \dots, b_{m-1})$
- $x^{\parallel n} = (x, \dots, x) \in X^n$

Kronecker Delta

- $\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

Lagrange Basis Polynomials

- $L_i = L_{\mathbb{D},i} \in \mathbb{F}[X]^{<N}$, $i \in \mathbb{N}_N$, $L_i(\omega^j) = \delta_{ij}$

1.2. Curves and Fields

- $\langle \omega \rangle = \mathbb{D} \subseteq \mathbb{F}^*$, $|\mathbb{D}| = N \in \mathbb{N}$
 - Cyclic subgroup generated by ω in the multiplicative group $\mathbb{F}^*$.

Elliptic Curves

- $J = J/\mathbb{F}$ – Twisted Edwards curve $ax^2 + y^2 = 1 + dx^2y^2$ defined over $\mathbb{F}$, with curve coefficient a .
- $\tilde{J} = J(\mathbb{F})$ – Group of $\mathbb{F}$ -rational points on J .
- $\mathbb{J} \subset \tilde{J}$ – Prime order subgroup of $\tilde{J}$.

Scalar Field

- $\mathbb{F}_{\mathbb{J}}$ – Field associated with the elliptic curve $\mathbb{J}$, with $|\mathbb{F}_{\mathbb{J}}| = |\mathbb{J}|$.
- $N_J = \lceil \log_2 |\mathbb{F}_{\mathbb{J}}| \rceil$ – Number of bits to represent an element of $\mathbb{F}_{\mathbb{J}}$.
- $N_K = N - N_J - 4$ – Maximum size of the ring handled with a domain of size N .

1.3. Support Functions

Unzip

- $\text{unzip} : \mathbb{J}^k \rightarrow (\mathbb{F}^k, \mathbb{F}^k); \quad \bar{p} \mapsto (\bar{p}_x, \bar{p}_y)$
 - Given a vector $\bar{p}$ of k elliptic curve points, unzip separates $\bar{p}$ into two vectors: $\bar{p}_x$ and $\bar{p}_y$, containing the x and y coordinates of each point, respectively.

Polynomial Interpolation

- $\text{Interpolate} : \mathbb{F}^k \rightarrow \mathbb{F}[x]^{<k}; \quad \bar{x} \mapsto f$
 - Construct a polynomial by interpolating the vector values over $\mathbb{D}$

Polynomial Commitment Scheme

- $\text{PCS.Commit} : \mathbb{F}[x] \rightarrow \mathbb{G}; \quad f \mapsto C_f$
 - Commits to a polynomial f over $\mathbb{F}$, with commitment in group $\mathbb{G}$.
- $\text{PCS.Open} : (\mathbb{G}, \mathbb{F}) \rightarrow (\mathbb{F}, \Pi); \quad (C_f, x) \mapsto (y, \pi)$
 - Evaluates the committed polynomial f at point x , returning evaluation y and proof π . The proof domain Π depends on the PCS.
- $\text{PCS.Verify} : (\mathbb{G}, \mathbb{F}, \mathbb{F}, \Pi) \rightarrow \mathbb{B}; \quad (C_f, x, y, \pi) \mapsto (0|1)$
 - Verifies whether $y = f(x)$ given the commitment C_f and proof π .

The reference implementation uses `fflonk`, a KZG variant that batches multiple polynomial openings into a single group element, reducing proof size and verification cost.

Fiat-Shamir Transform

- $\text{FS} : \mathbb{S} \rightarrow \mathbb{F}; \mathbf{s} \mapsto x$
 - Maps a serializable object $\mathbf{s} \in \mathbb{S}$ to $\mathbb{F}$, typically via some cryptographically secure hash function.

The reference implementation uses **ark-transcript**, a Merlin-like transcript based on SHAKE128 that appends the length of the serialized object (rather than prepending it as Merlin does).

2. Parameters

2.1. Scheme Specific

- $\square \in \mathbb{J}$ – Padding element, a point on $\mathbb{J}$ with unknown discrete logarithm.
- $H \in \mathbb{J}$ – Pedersen blinding base point.
- $\overline{H} = (H, 2H, 4H, \dots, 2^{N_J-1}H) \in \mathbb{J}^{N_J}$ – Vector of scaled multiples of H .
- $S \in \mathbb{J}$ – Seed point for accumulation, with unknown discrete logarithm. Choosing $S \in \mathbb{J} \setminus \mathbb{J}$ is strongly recommended as it guarantees the accumulator never reaches the identity element.

2.2. Public Data

- $\overline{PK} \in \mathbb{J}^{N_K}$ – Vector of public keys in the ring, padded with $\square$ to length N_K if needed.

2.3. Witness Data

- $t \in \mathbb{F}_{\mathbb{J}}$ – Prover’s one-time secret.
- $k \in \mathbb{N}_{N_K}$ – Prover’s index within the ring, identifying which public key in $\overline{PK}$ belongs to the prover.

2.4. Preprocessing

2.4.1. Public Input Preprocessing Concatenate ring points with scaled multiples of H :

$$\overline{P} = \overline{PK} \parallel \overline{H} = (P_0, \dots, P_{N-5}) \in \mathbb{J}^{N-4}$$

The $N - 4$ point entries correspond to the $N - 4$ constrained accumulator transitions. The coordinate vectors are padded with 4 trailing zeros to fill the domain:

$$\begin{aligned} \overline{p}_x &= (P_{x,0}, \dots, P_{x,N-5}, 0, 0, 0, 0) \in \mathbb{F}^N \\ \overline{p}_y &= (P_{y,0}, \dots, P_{y,N-5}, 0, 0, 0, 0) \in \mathbb{F}^N \end{aligned}$$

Ring items selector:

$$\bar{\sigma} = 1^{\|N_K\|} \parallel 0^{\|N-N_K\|} \in \mathbb{F}^N$$

2.4.2. Interpolation The resulting vectors are interpolated over $\mathbb{D}$:

$$p_x = \text{Interpolate}(\bar{p}_x)$$

$$p_y = \text{Interpolate}(\bar{p}_y)$$

$$\sigma = \text{Interpolate}(\bar{\sigma})$$

2.4.3. Commit to the constructed vectors

$$C_{p_x} = \text{PCS.Commit}(p_x)$$

$$C_{p_y} = \text{PCS.Commit}(p_y)$$

$$C_{\sigma} = \text{PCS.Commit}(\sigma)$$

2.5. Relation to Prove

Knowledge of k and t such that $R = PK_k + tH$.

$$\mathfrak{R}_H = \{(R, \overline{PK}; k, t) \mid R = PK_k + tH; R \in \mathbb{J}, \overline{PK} \in \mathbb{J}^{N_K}, k \in \mathbb{N}_{N_K}, t \in \mathbb{F}_{\mathbb{J}}\}$$

3. Prover

3.1. Witness Polynomials

3.1.1. Bits Vector

- $\bar{k} \in \mathbb{B}^{N_K}$ – Binary vector representing the index k in the ring. $\bar{k}$ has N_K elements where $k_i = \delta_{ik}$
- $\bar{t} \in \mathbb{B}^{N_J}$ – Binary representation of the secret scalar t , with t_i representing the i -th bit of t in little-endian order, i.e., $t = \sum t_i 2^i$, for $i \in \mathbb{N}_{N_J}$

The bits vector $\bar{b}$ is constructed by concatenating $\bar{k}$ and $\bar{t}$, followed by a single 0.

$$\bar{b} = \bar{k} \parallel \bar{t} \parallel (0)$$

3.1.2. Conditional Sum Accumulator Vectors

$$ACC_0 = S, \quad ACC_i = ACC_{i-1} + b_{i-1} P_{i-1}, \quad i = 1, \dots, N-4$$

- The accumulator is initialized with the seed point S .
- The accumulator is updated at each index i based on the previous value and the product of b_{i-1} and P_{i-1} .

The resulting accumulator points are finally separated into x and y coordinates:

$$(\overline{acc}_x, \overline{acc}_y) = \text{unzip}(\overline{ACC})$$

3.1.3. Inner Product Accumulator Vector

$$acc_{ip_0} = 0, \quad acc_{ip_i} = acc_{ip_{i-1}} + b_{i-1}\sigma_{i-1}, \quad i = 1, \dots, N-4$$

- The accumulator is initialized with 0.
- The accumulator is updated at each index i based on the previous value and the product of b_{i-1} and s_{i-1}

3.1.4. Interpolation and Commitments In hiding (zero-knowledge) mode, the resulting vectors are interpolated over $\mathbb{D}$ with random values $\{r_i\}$ occupying the last 3 domain positions ($N-3, N-2, N-1$). These positions are unconstrained and serve to randomize the witness polynomials, preventing information leakage through polynomial commitments.

$$\begin{aligned} b &= \text{Interpolate}(\bar{b} \parallel (r_1, r_2, r_3)) \\ acc_x &= \text{Interpolate}(\overline{acc_x} \parallel (r_4, r_5, r_6)) \\ acc_y &= \text{Interpolate}(\overline{acc_y} \parallel (r_7, r_8, r_9)) \\ acc_{ip} &= \text{Interpolate}(\overline{acc_{ip}} \parallel (r_{10}, r_{11}, r_{12})) \end{aligned}$$

Commit to the witness derived polynomials:

$$\begin{aligned} C_b &= \text{PCS.Commit}(b) \\ C_{acc_{ip}} &= \text{PCS.Commit}(acc_{ip}) \\ C_{acc_x} &= \text{PCS.Commit}(acc_x) \\ C_{acc_y} &= \text{PCS.Commit}(acc_y) \end{aligned}$$

3.2. Constraints

Constraints are polynomials constructed to evaluate to zero when satisfied; a non-zero evaluation indicates a violation.

Note. When evaluating a polynomial f at $x = \omega^k \in \mathbb{D}$ for some $k \in \mathbb{N}$, $f(\omega x)$ gives the value of the polynomial at the next position in the evaluation domain ($\omega x = \omega^{k+1}$).

3.2.1. Inner Product

$$c_1(x) = (acc_{ip}(\omega x) - acc_{ip}(x) - b(x)\sigma(x))(x - \omega^{N-4})$$

This constraint ensures the inner product accumulator $acc_{ip}(x)$ is correctly updated, satisfying $acc_{ip}(\omega x) = acc_{ip}(x) + b(x)\sigma(x)$.

The factor $(x - \omega^{N-4})$ ensures the constraint holds at all points including $x = \omega^{N-4}$, where $c_1(x)$ automatically vanishes.

3.2.2. Conditional Addition

$$\begin{aligned}
c_2(x) = & \left(b(x) \left(acc_x(\omega x) (acc_y(x) \cdot p_y(x) + a \cdot acc_x(x) \cdot p_x(x)) \right. \right. \\
& \left. \left. - acc_x(x) \cdot acc_y(x) - p_x(x) \cdot p_y(x) \right) \right. \\
& \left. + (1 - b(x)) (acc_x(\omega x) - acc_x(x)) \right) \times (x - \omega^{N-4}) \\
c_3(x) = & \left(b(x) \left(acc_y(\omega x) (acc_x(x) \cdot p_y(x) - p_x(x) \cdot acc_y(x)) \right. \right. \\
& \left. \left. - acc_x(x) \cdot acc_y(x) + p_x(x) \cdot p_y(x) \right) \right. \\
& \left. + (1 - b(x)) (acc_y(\omega x) - acc_y(x)) \right) \times (x - \omega^{N-4})
\end{aligned}$$

These constraints enforce correct elliptic curve addition for the x and y components, respectively, controlled by the Boolean variable $b(x)$:

- **When $b(x) = 1$:** $c_2(x)$ and $c_3(x)$ enforce the correct elliptic curve addition for the x and y components, respectively.
- **When $b(x) = 0$:** both constraints ensure the accumulator remains unchanged.

The factor $(x - \omega^{N-4})$ nullifies the constraint at $x = \omega^{N-4}$, where it does not apply.

3.2.3. Booleanity

$$c_4(x) = b(x)(1 - b(x))$$

Ensures that the polynomial $b(x)$ acts as a Boolean variable, taking only values 0 or 1.

- **If $b(x) = 0$ or $b(x) = 1$,** then $c_4(x) = 0$.
- **If $b(x)$ takes any value other than 0 or 1,** $c_4(x)$ will be non-zero, violating the constraint.

3.2.4. Conditional Addition Boundary Given the seed point $S = (s_x, s_y)$ and the expected result $R = (r_x, r_y)$, the verifier computes $E = S + R$ (EC point addition) with $E = (e_x, e_y)$. The constraints are:

$$\begin{aligned}
c_5(x) = & (acc_x(x) - s_x)L_0(x) + (acc_x(x) - e_x)L_{N-4}(x) \\
c_6(x) = & (acc_y(x) - s_y)L_0(x) + (acc_y(x) - e_y)L_{N-4}(x)
\end{aligned}$$

These constraints ensure the accumulator components take specific values at the conditional addition boundaries:

- **At** $x = \omega^0$: $L_0(x) = 1$ and $L_{N-4}(x) = 0$, enforcing $acc_x(\omega^0) = s_x$ and $acc_y(\omega^0) = s_y$.
- **At** $x = \omega^{N-4}$: $L_0(x) = 0$ and $L_{N-4}(x) = 1$, enforcing $acc_x(\omega^{N-4}) = e_x$ and $acc_y(\omega^{N-4}) = e_y$.

3.2.5. Inner Product Boundary

$$c_7(x) = acc_{ip}(x)L_0(x) + (acc_{ip}(x) - 1)L_{N-4}(x)$$

This constraint ensure the accumulator components take specific values at the conditional addition boundaries:

- **At** $x = \omega^0$: $L_0(x) = 1$ and $L_{N-4}(x) = 0$, enforcing $acc_{ip}(\omega^0) = 0$.
- **At** $x = \omega^{N-4}$: $L_0(x) = 0$ and $L_{N-4}(x) = 1$, enforcing $acc_{ip}(\omega^{N-4}) = 1$.

3.3. Constraints Aggregation

3.3.1. Aggregation Polynomial The protocol aggregates all constraints into a single polynomial for efficiency.

Using the Fiat-Shamir heuristic, sample the aggregation coefficients:

$$\{\alpha_i\}_{i=1}^7 \leftarrow \text{FS}(C_b, C_{acc_{ip}}, C_{acc_x}, C_{acc_y})$$

Construct the aggregated polynomial:

$$c(x) = \left(\sum_{i=1}^7 \alpha_i c_i(x) \right) \cdot \prod_{k=1}^3 (x - \omega^{N-k})$$

In hiding mode, the factor $\prod_{k=1}^3 (x - \omega^{N-k})$ zeros out the aggregated constraints at the 3 randomized domain positions, ensuring the quotient polynomial $q(x)$ is well-formed despite these unconstrained rows. This factor is not present in non-hiding mode.

3.3.2. Quotient Polynomial The quotient polynomial is computed as:

$$q(x) = \frac{c(x)}{x^N - 1}$$

Dividing by $X^N - 1$ ensures that the aggregated constraints encoded in $c(x)$ are enforced consistently across the entire evaluation domain $\mathbb{D}$ while reducing the degree of the polynomial.

3.3.3. Quotient Polynomial Commitment and Challenge The prover commits to the quotient polynomial q :

$$C_q = \text{PCS.Commit}(q)$$

The prover receives the evaluation point ζ in response:

$$\zeta \leftarrow \text{FS}(C_q)$$

3.3.4. Relevant Polynomials Evaluation Evaluate the relevant polynomials at the sampled evaluation point ζ :

$$\begin{aligned} p_{x,\zeta} &= p_x(\zeta) \\ p_{y,\zeta} &= p_y(\zeta) \\ \sigma_\zeta &= \sigma(\zeta) \\ b_\zeta &= b(\zeta) \\ acc_{ip,\zeta} &= acc_{ip}(\zeta) \\ acc_{x,\zeta} &= acc_x(\zeta) \\ acc_{y,\zeta} &= acc_y(\zeta) \end{aligned}$$

3.3.5. Linearization Polynomial The linearization polynomials are constructed to enable the verifier to evaluate certain parts of the constraint polynomials at $\zeta\omega$ while independently reconstructing the evaluation of q at ζ using the “relevant polynomial evaluations” provided by the prover as part of the proof.

In particular we require these contributions just for the accumulators constraints.

Accumulator inner product (c_1) contribution:

$$l_1(x) = (\zeta - \omega^{N-4}) acc_{ip}(x)$$

Conditional addition accumulators ($c_{2,3}$) contributions:

$$\begin{aligned} l_2(x) &= (\zeta - \omega^{N-4}) (b_\zeta (acc_{y,\zeta} \cdot p_{y,\zeta} + a \cdot acc_{x,\zeta} \cdot p_{x,\zeta}) + 1 - b_\zeta) acc_x(x) \\ l_3(x) &= (\zeta - \omega^{N-4}) (b_\zeta (acc_{x,\zeta} \cdot p_{y,\zeta} - p_{x,\zeta} \cdot acc_{y,\zeta}) + 1 - b_\zeta) acc_y(x) \end{aligned}$$

Linearized constraints are aggregated using $\{\alpha_i\}$ coefficients and evaluated at $\zeta\omega$:

$$\begin{aligned} l(x) &= \sum_{i=1}^3 \alpha_i l_i(x) \\ l_{\zeta\omega} &= l(\zeta\omega) \end{aligned}$$

3.3.6. Sample Aggregation Coefficients Sample the aggregation coefficients $\{\nu_i\}$ using the Fiat-Shamir heuristic and compute the aggregate polynomial agg :

$$\{\nu_i\}_{i=1}^8 \leftarrow \text{FS}(p_{x,\zeta}, p_{y,\zeta}, \sigma_\zeta, b_\zeta, acc_{ip,\zeta}, acc_{x,\zeta}, acc_{y,\zeta}, l_{\zeta\omega})$$

Construct the aggregate polynomial:

$$agg(x) = \nu_1 p_x(x) + \nu_2 p_y(x) + \nu_3 \sigma(x) + \nu_4 b(x) + \nu_5 acc_{ip}(x) + \nu_6 acc_x(x) + \nu_7 acc_y(x) + \nu_8 q(x)$$

3.3.7. Proof Construction Open the aggregate polynomial agg at ζ and the linearization polynomial l at $\zeta\omega$:

$$\Pi_\zeta = \text{PCS.Open}(agg, \zeta)$$

$$\Pi_{\zeta\omega} = \text{PCS.Open}(l, \zeta\omega)$$

Construct the proof as follows:

$$\Pi = (C_b, C_{acc_{ip}}, C_{acc_x}, C_{acc_y}, p_{x,\zeta}, p_{y,\zeta}, \sigma_\zeta, b_\zeta, acc_{ip,\zeta}, acc_{x,\zeta}, acc_{y,\zeta}, C_q, l_{\zeta\omega}, \Pi_\zeta, \Pi_{\zeta\omega})$$

4. Verifier

4.1. Inputs

Commitments to the ring public keys and the selector, prepared during the pre-processing phase:

$$(C_{p_x}, C_{p_y}, C_\sigma)$$

The claimed result point, allegedly $R = PK_k + tH$ for some k and t known to the prover:

$$R = (r_x, r_y)$$

Proof which contains all the necessary commitments, evaluations, and openings needed for the verifier to perform the validation checks:

$$\Pi = (C_b, C_{acc_{ip}}, C_{acc_x}, C_{acc_y}, p_{x,\zeta}, p_{y,\zeta}, \sigma_\zeta, b_\zeta, acc_{ip,\zeta}, acc_{x,\zeta}, acc_{y,\zeta}, C_q, l_{\zeta\omega}, \Pi_\zeta, \Pi_{\zeta\omega})$$

4.2. Verification

4.2.1. Fiat-Shamir Challenges Recovery of aggregation coefficients and evaluation point:

$$\{\alpha_i\}_{i=1}^7 \leftarrow \text{FS}(C_b, C_{acc_{ip}}, C_{acc_x}, C_{acc_y})$$

$$\zeta \leftarrow \text{FS}(C_q)$$

$$\{\nu_i\}_{i=1}^8 \leftarrow \text{FS}(p_{x,\zeta}, p_{y,\zeta}, \sigma_\zeta, b_\zeta, acc_{ip,\zeta}, acc_{x,\zeta}, acc_{y,\zeta}, l_{\zeta\omega})$$

4.2.2. Contributions to the Constraints Evaluated at ζ The following expressions represent the contributions to the constraint polynomials evaluated at the point ζ :

$$\begin{aligned}\tilde{c}_{1,\zeta} &= -(acc_{ip,\zeta} + b_\zeta \sigma_\zeta)(\zeta - \omega^{N-4}) \\ \tilde{c}_{2,\zeta} &= \{-b_\zeta(acc_{x,\zeta} \cdot acc_{y,\zeta} + p_{x,\zeta} \cdot p_{y,\zeta}) - (1 - b_\zeta)acc_{x,\zeta}\}(\zeta - \omega^{N-4}) \\ \tilde{c}_{3,\zeta} &= \{-b_\zeta(acc_{x,\zeta} \cdot acc_{y,\zeta} - p_{x,\zeta} \cdot p_{y,\zeta}) - (1 - b_\zeta)acc_{y,\zeta}\}(\zeta - \omega^{N-4}) \\ c_4 &= b_\zeta(1 - b_\zeta) \\ c_5 &= (acc_{x,\zeta} - s_x)L_0(\zeta) + (acc_{x,\zeta} - e_x)L_{N-4}(\zeta) \\ c_6 &= (acc_{y,\zeta} - s_y)L_0(\zeta) + (acc_{y,\zeta} - e_y)L_{N-4}(\zeta) \\ c_7 &= acc_{ip,\zeta}L_0(\zeta) + (acc_{ip,\zeta} - 1)L_{N-4}(\zeta)\end{aligned}$$

Note: The tilde ($\sim$) above the first three polynomials indicates that these are only partial contributions, representing the components evaluated at ζ . The components evaluated at $\zeta\omega$ are added later by the linearization aggregated polynomial found within the proof ($l_{\zeta\omega}$).

4.2.3. Evaluation of the Quotient Polynomial at ζ Aggregate the contributions along with the linearization polynomial evaluated at $\zeta\omega$ to compute the evaluation of the quotient polynomial at ζ :

$$q_\zeta = \frac{(\sum_{i=1}^7 \alpha_i c_i + l_{\zeta\omega}) \prod_{k=1}^3 (\zeta - \omega^{N-k})}{\zeta^N - 1}$$

Compute the aggregate commitment C_{agg} using the aggregation coefficients ν_i :

$$C_{agg} = \nu_1 C_{p_x} + \nu_2 C_{p_y} + \nu_3 C_\sigma + \nu_4 C_b + \nu_5 C_{acc_{ip}} + \nu_6 C_{acc_x} + \nu_7 C_{acc_y} + \nu_8 C_q$$

Compute the aggregate evaluation agg_ζ using the same coefficients:

$$agg_\zeta = \nu_1 p_{x,\zeta} + \nu_2 p_{y,\zeta} + \nu_3 \sigma_\zeta + \nu_4 b_\zeta + \nu_5 acc_{ip,\zeta} + \nu_6 acc_{x,\zeta} + \nu_7 acc_{y,\zeta} + \nu_8 q_\zeta$$

Verify the aggregate polynomial opening at ζ using Π_ζ :

$$\text{PCS.Verify}(C_{agg}, \zeta, agg_\zeta, \Pi_\zeta)$$

4.2.4. Evaluation of the Linearization Polynomial at $\zeta\omega$ Compute the individual linearization polynomial commitments:

$$\begin{aligned}C_{l_1} &= (\zeta - \omega^{N-4})C_{acc_{ip}} \\ C_{l_2} &= (\zeta - \omega^{N-4})(b_\zeta(acc_{y,\zeta} \cdot p_{y,\zeta} + a \cdot acc_{x,\zeta} \cdot p_{x,\zeta}) + 1 - b_\zeta)C_{acc_x} \\ C_{l_3} &= (\zeta - \omega^{N-4})(b_\zeta(acc_{x,\zeta} \cdot p_{y,\zeta} - p_{x,\zeta} \cdot acc_{y,\zeta}) + 1 - b_\zeta)C_{acc_y}\end{aligned}$$

Aggregate the linearization polynomial commitments using $\{\alpha_i\}$ coefficients.

$$C_l = \sum_{i=1}^3 \alpha_i C_{l_i}$$

Verify the aggregate linearization polynomial opening at $\zeta\omega$ using $\Pi_{\zeta\omega}$:

$$\text{PCS.Verify}(C_l, \zeta\omega, l_{\zeta\omega}, \Pi_{\zeta\omega})$$

5. Acknowledgements

This specification is primarily derived from Sergey Vasilyev’s original writeup and reference implementation, as cited in the references.

6. References

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4. ark-transcript (Fiat-Shamir): <https://crates.io/crates/ark-transcript>